

Aishik Mandal

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📄 Aishik Mandal

Education

- **Technical University of Darmstadt**
PhD in Computer Science, 2023-
- **Indian Institute of Technology, Kharagpur** CGPA: 9.61/10
Bachelor of Technology in Electronics and Electrical Communication Engineering, 2018-2023
Master of Technology in Artificial Intelligence, Machine Learning and Applications
Minor in Computer Science and Engineering
Micro Specialisation in Embedded Control, Software, Modelling and Design

Internships

- **Multimodal Turn Taking in Conversational Agents** INRIA Paris, France
Research Intern at [Articulabo, Cognitive Machine Learning Group](#) under Prof. Justine Cassell, May 2022-Dec 2022
- **Revenue Function of Hierarchical Clustering in Comparison Framework** TU Munich, Germany
Research Intern at [Theoretical Foundations of AI](#) under Prof. Debarghya Ghoshdastidar, Jun 2021-Aug 2021

Thesis

- **Universal Transformer for Multimodal Self-supervised Learning** IIT Kharagpur
Master's Thesis, under Prof. Jiaul Hoque Paik, [Centre of Excellence in AI\(CoEAI\)](#) Aug 2022-Apr 2023
 - Proposed a **novel loss function** to train a transformer which can **encode images and texts in a shared embedding space**.
 - **Trained a vanilla transformer encoder** from scratch with the proposed loss function on CoCo image captioning dataset.
 - **Evaluated** the performance of the Universal Transformer in **multimodal and single modality situations**.
 - Conducted **error analysis** and suggested **adversarial training** to achieve improvements.
- **Discourse Mutual Information for Dialogue Understanding and Response Generation** IIT Kharagpur
Bachelor's Thesis, under Prof. Pawan Goyal, [Department of Computer Science and Engineering](#) Dec 2020-Apr 2021
 - Performed extensive experimentation on a transformer based dual encoder architecture to encode context and response in a dialog, with the purpose of increasing the proposed **Discourse Mutual Information** objective function.
 - Performed various downstream **dialog-understanding tasks** as a means of evaluating the representations learned.
 - Achieved an accuracy improvement of upto **5.8%** on **dialog classification tasks** and upto **15.3%** on **dialog evaluation tasks** over state of the art baselines while performing at par with the baselines on response selection tasks.
 - Designed and trained a **decoder** to obtain responses from response encoding and proposed **algorithms** to map context encoding to response encoding.
 - Performed **exploratory analysis** to understand the **features captured** by the encoder.

Publications

- **MAGneT: Coordinated Multi-Agent Generation of Synthetic Multi-Turn Mental Health Counseling Sessions**
Aishik Mandal, Tanmoy Chakraborty, Iryna Gurevych, *preprint* ([Link to the paper](#))
- **A Comprehensive Survey of Datasets for Clinical Mental Health AI Systems**
Aishik Mandal, Prottay Kumar Adhikary, Hiba Arnaout, Iryna Gurevych, Tanmoy Chakraborty, *preprint* ([Link to the paper](#))
- **CaMMT: Benchmarking Culturally Aware Multimodal Machine Translation**
Emilio Villa-Cueva, Sholpan Bolatzhanova, Diana Turmakhan, Kareem Elzeky, ..., **Aishik Mandal**, ..., Injy Hamed, Atnafu Lambebo Tonja, Tamar Solorio, *preprint* ([Link to the paper](#))
- **Towards Privacy-aware Mental Health AI Models: Advances, Challenges, and Opportunities**
Aishik Mandal, Tanmoy Chakraborty, Iryna Gurevych, *Nature Computational Science* ([Link to the paper](#))

- **Enhancing Depression Detection via Question-wise Modality Fusion**
Aishik Mandal, Dana Atzil-Slonim, Tamar Solorio, Iryna Gurevych, *Oral, Workshop on Computational Linguistics and Clinical Psychology (CLPsych 2025)* ([Link to the paper](#))([Code-base Link](#))
- **CVQA: Culturally-diverse Multilingual Visual Question Answering Benchmark**
David Orlando Romero Mogrovejo, Chenyang Lyu, Haryo Akbarianto Wibowo, Santiago Góngora, **Aishik Mandal**,...,Tamar Solorio, Alham Fikri Aji, *Oral, Neural Information Processing Systems Datasets and Benchmarks Track (NEURIPS 2024)* ([Link to the paper](#))
- **A Revenue Function for Comparison-Based Hierarchical Clustering**
Aishik Mandal, Michaël Perrot, Debarghya Ghoshdastidar, *Transactions on Machine Learning Research (TMLR, March 2023)* ([Link to the paper](#))([Code-base Link](#))
- **Representation Learning for Conversational Data using Discourse Mutual Information Maximization**
Bishal Santra, Sumegh Roychowdhury, **Aishik Mandal**, Vasu Gurram, Atharva Naik, Manish Gupta, Pawan Goyal, *2022 Annual Conference of the North American Chapter of the Association for Computational Linguistics (NAACL 2022)* ([Link to the paper](#)) ([Link to project webpage](#))
- **Knowledge-Aware Neural Networks for Medical Forum Question Classification**
Soumyadeep Roy, Sudip Chakraborty, **Aishik Mandal**, Gunjan Balde, Prakhar Sharma, Anandhavelu Natrajan, Megha Khosla, Shamik Sural, Niloy Ganguly, *30th ACM International Conference on Information and Knowledge Management (CIKM 2021)* ([Link to the paper](#)) ([Code-base Link](#))

Skills and Expertise

- **Programming Languages and Softwares:** Python / C++ / C / MATLAB / OpenSmile / OpenFace / Elan
- **Libraries:** PyTorch / DeepSpeed / vLLM / ScikitLearn / Pandas / NumPy

Student Supervision

- **Masters Thesis, TU Darmstadt**
 - Mateusz Ciesielski (Apr 2025-Oct 2025): Predicting Personality Domains: A Multimodal Approach
 - Paulina Kuczykowska (Sep 2023-Feb 2024): Predicting Depression Severity from Therapy Dialogues

Teaching

- **Teaching Assistant, TU Darmstadt(2025):** Ethics for Natural Language Processing.
- **Teaching Assistant, TU Darmstadt(2024):** Ethics for Natural Language Processing.
- **Teaching Assistant, IIT Kharagpur(2023):** Big Data Processing.

Reviewing

- Transactions on Machine Learning Research (TMLR 2025)
- AAAI AI for Social Impact Track 2026

Awards And Achievements

- **ELIZA Scholarship:** Received ELIZA Scholarship for ELLIS exchange at INRIA Montpellier (2025).
- **Silver Medalist:** Received Silver Medal for holding department rank 1 in Electronics and Electrical Communication Engineering at IIT Kharagpur (2023).
- **Charpak Lab Scholarship Awardee:** Received Charpak Lab scholarship for summer internship at INRIA (2022).
- **GKF International Internship Scholarship:** Received Guru Krupa IITKGP Foundation Scholarship from IITKGP Foundation for summer internship at INRIA (2022).
- **DAAD WISE Awardee:** Received WISE scholarship for summer internship at TU Munich (2021).

Conferences

- **CLPsych Workshop, NAACL 2025**
- **Google Research Week, Google Research India(2023)**